

 SMART AGRICULTURE AI

# AquaWise

“Aqua” for vital water. “Wise” for intelligent machine-learning decisions.

 Predictive crop hydration with zero unnecessary waste

## CORE OBJECTIVE

# Right Water. Right Time.

## The AquaWise Mandate

Give crops the water they need while reducing water waste.

### Precision

Exact dosage delivery.

### Efficiency

Zero runoff loss.



# Three Interconnected Domains



## Artificial Intelligence

Predictive machine learning algorithms.



## Smart Agriculture

Crop-specific growth requirements.



## Water Conservation

Solving real-world freshwater scarcity.

# Multi-Factor Data Pipeline



## Soil Moisture Levels

Real-time root hydration monitoring.



## Temperature & Humidity

Vapor deficit calculation factors.



## Weather & Rainfall Forecasts

Preventing irrigation before storms.



## Crop Profiling

Species-dependent water thresholds.

# Comparing Model Architecture

## Random Forest

**Ensemble:** Parallel decision trees.

**Stability:** High noise tolerance.

**Output:** Dual threshold classification.

## XGBoost

**Gradient:** Sequential error correction.

**Accuracy:** Sharp boundary optimization.

**Volume:** Water quantity estimation.

FIELD EXECUTION

# Intelligent Field Decisions

DECISION: DELAY

## High Moisture + Incoming Rain

System delays irrigation cycle.

Result: Zero wasted electricity & water.

DECISION: ACTIVATE

## Dry Root Zone + Peak Heat

System activates calculated irrigation.

Result: Prevents drought stress instantly.



# Smarter Farming for a Thirsty Planet

AquaWise turns environmental data into precision action.

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THANK YOU • QUESTIONS & DISCUSSION

## Image Sources



[https://img.magnific.com/premium-photo/dynamic-center-pivot-irrigation-system-watering-lush-green-crops-golden-hour-sunset-with-high-quality-professional-line-art-symbols-modern-creative-design-digital-branding-projects\\_538547-20918.jpg?semt=ais\\_hybrid&w=740&q=80](https://img.magnific.com/premium-photo/dynamic-center-pivot-irrigation-system-watering-lush-green-crops-golden-hour-sunset-with-high-quality-professional-line-art-symbols-modern-creative-design-digital-branding-projects_538547-20918.jpg?semt=ais_hybrid&w=740&q=80)

Source: [www.magnific.com](http://www.magnific.com)

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<https://eu-images.contentstack.com/v3/assets/bltdd43779342bd9107/blt6b5b1f2876a08f4e/6409095ce5335a38e1139427/0313W2-3443-1800x1012.jpg>

Source: [www.farmprogress.com](http://www.farmprogress.com)